

DeepThinkTrader v3.0 Setup Guide

Version: 3.0 (Execution Intelligence) | **Generated:** March 22, 2026

CRITICAL WARNING: Start with paper trading ONLY. Never use real money until you have backtested for 3+ months. Trading bots can lose 100% of capital. This project is educational and experimental.

1. Required API Keys

You need three free API accounts. All keys go in the `.env` file (never commit this file).

1.1 Alpaca Markets (Paper Trading)

1. Go to **alpaca.markets** and create a free account
2. Navigate to **Paper Trading** dashboard
3. Click **API Keys** → **Generate New Key**
4. Copy both the **API Key ID** and **Secret Key** (secret shown only once!)

Variable	Description	Example
ALPACA_API_KEY	Your Alpaca API Key ID	PK... (20 chars)
ALPACA_SECRET_KEY	Your Alpaca Secret Key	abc123... (40 chars)
ALPACA_BASE_URL	Paper trading endpoint	https://paper-api.alpaca.markets

1.2 NewsAPI

1. Go to **newsapi.org** and register for free
2. Copy your API key from the dashboard

3. Free tier: 500 requests/day (sufficient for hourly research on 5 tickers)

1.3 Reddit API (PRAW) — Optional

1. Log into Reddit, go to **reddit.com/prefs/apps**
2. Click "**create another app...**" at the bottom
3. Select "**script**" type, name: `DeepThinkTrader` , redirect: `http://localhost:8080`
4. Note the **client ID** (under the app name) and **secret**

1.4 RapidAPI (Twelve Data) — Optional

1. Go to **rapidapi.com** and create a free account
2. Subscribe to **Twelve Data** API (free tier: 800 req/day)
3. Copy your RapidAPI key

2. Trading Parameters

2.1 Trade Modes

Set `TRADE_MODE=safe|normal|aggressive` in `.env` or switch via dashboard.

Parameter	Safe	Normal	Aggressive
Max Risk/Trade	1%	2%	3%
Daily Loss Limit	3%	5%	8%
Min Conviction	9.0	7.5	6.0
Min R:R Ratio	3:1	2:1	1.5:1
Max Position %	5%	10%	15%
Max Positions	5	10	15

2.2 Risk-First Parameters

Parameter	Variable	Default	Description
Kelly Safety	KELLY_SAFETY_MULTIPLIER	0.5	Half-Kelly for conservative sizing
Drawdown Halt	MAX_DRAWDOWN_HALT_PCT	0.08 (8%)	Block all entries if drawdown exceeds this
Volatility Mult	VOLATILITY_ATR_MULTIPLIER	3.0	Cut risk 50% if ATR > 3x median (real 30-day)
Min ADV Ratio	MIN_ADV_RATIO	5	Shares must be < ADV/5
Max Risk of Ruin	MAX_RISK_OF_RUIN_PCT	0.01 (1%)	Block if RoR exceeds 1%
Min Edges	MIN_EDGES_REQUIRED	2	Require 2/3 edges (Fund+Tech+Sent)
SPY Circuit Breaker	CIRCUIT_BREAKER_SPY_DROP_PCT	-2.0	Block longs if SPY down > 2%

VIX Circuit Breaker	CIRCUIT_BREAKER_VIX_THRESHOLD	30	Block ALL entries when VIX $\geq$ 30
Earnings Exit	EARNINGS_EXIT_DAYS	2	Auto-close within 2 days / 12 hours of earnings

2.3 Execution Quality Parameters (v3.0)

Parameter	Variable	Default	Description
Max Spread	MAX_SPREAD_PCT	1.0%	Block market orders with wide spreads
Penny Max Spread	PENNY_MAX_SPREAD_PCT	2.0%	Higher spread tolerance for penny stocks
Sector Limit	MAX_SECTOR_EXPOSURE_PCT	25%	Max portfolio % per GICS sector
Gap Risk ATR	GAP_RISK_ATR_THRESHOLD	5.0%	ATR% triggering gap risk reduction
Gap Reduction	GAP_RISK_POSITION_REDUCTION	50%	Position size multiplier for high gap risk
Obsidian Vault	OBSIDIAN_VAULT_PATH	~/Documents/RHVault/RHVault	Path to Obsidian vault for SA emails

2.3 Exit Management Parameters

Parameter	Variable	Default	Description
Exit Check Interval	EXIT_CHECK_INTERVAL_MINUTES	5	Fast price checks every 5 min
Trail Activation	TRAILING_STOP_ACTIVATION_PCT	2.0%	Switch to trailing stop at 2% profit
Trail Distance (Main)	TRAILING_STOP_DISTANCE_PCT	1.5%	Trail distance for main portfolio

Trail Distance (Penny)	PENNY_TRAILING_STOP_DISTANCE_PCT	3.0%	Wider trail for volatile penny stocks
Scale-Out	SCALE_OUT_ENABLED	true	Sell 33% at 1R, 33% at 2R
Time Stop	TIME_STOP_DAYS	15	Auto-exit stale positions after 15 days
Penny Limit Slip	PENNY_LIMIT_SLIPPAGE_PCT	0.5%	Limit order buffer for penny stocks

3. Risk Management (15 Pre-Trade Checks)

These safety checks run BEFORE every trade and cannot be overridden via configuration.

#	Check	Action if Failed
1	Conviction meets threshold	HOLD
2	Risk within limit	BLOCKED
3	Reward:Risk ratio acceptable	BLOCKED
4	Daily loss limit OK	BLOCKED for rest of day
5	Open position count OK	BLOCKED
6	No duplicate position	BLOCKED
7	Market hours (9:30-4:00 ET)	BLOCKED
8	Drawdown from peak < 8%	BLOCKED (all new entries)
9	Risk of ruin < 1%	BLOCKED
10	Liquidity: shares < ADV/5	Auto-reduce or BLOCKED
11	Multi-edge: 2/3 edges firing	HOLD
12	Market health: SPY > 2% OR VIX ≥ 30	BLOCKED (SPY=longs, VIX=all)
13	No earnings within 2 days / 12 hours	Auto-close or BLOCKED
14	Bid-ask spread acceptable	BLOCKED
15	Sector concentration < 25%	BLOCKED

4. Quick Setup Checklist

#	Step	Done?
1	Create Alpaca paper account and get API keys	☐

2	Get NewsAPI key (free tier)	☐
3	Create Reddit app and get PRAW credentials (optional)	☐
4	Get RapidAPI key for Twelve Data (optional)	☐
5	Copy <code>.env.template</code> to <code>.env</code> and fill in keys	☐
6	Create Python venv and install requirements	☐
7	Run <code>python main.py</code> once to test single cycle	☐
8	Run <code>python main.py ticker NVDA</code> to test edge validation	☐
9	Run <code>streamlit run dashboard.py</code> to verify dashboard	☐
10	Start scheduled loop: <code>python main.py</code>	☐

5. .env File Template

```
# Trade Mode: safe, normal, aggressive
TRADE_MODE=normal

# Alpaca API (Paper Trading)
ALPACA_API_KEY=your_alpaca_api_key
ALPACA_SECRET_KEY=your_alpaca_secret_key
ALPACA_BASE_URL=https://paper-api.alpaca.markets

# NewsAPI
NEWSAPI_KEY=your_newsapi_key

# RapidAPI (Twelve Data - optional)
RAPIDAPI_KEY=your_rapidapi_key

# Reddit PRAW (optional)
REDDIT_CLIENT_ID=your_reddit_client_id
REDDIT_CLIENT_SECRET=your_reddit_client_secret
REDDIT_USER_AGENT=DeepThinkTrader/1.0

# Penny Portfolio
PENNY_ENABLED=true

# Override defaults (optional)
# EXIT_CHECK_INTERVAL_MINUTES=5
# TRAILING_STOP_ACTIVATION_PCT=2.0
# MIN_EDGES_REQUIRED=2
```

```
# CIRCUIT_BREAKER_SPY_DROP_PCT=-2.0  
# EARNINGS_EXIT_DAYS=2
```

DeepThinkTrader v3.0 — Execution Intelligence — Paper Trading Mode Only

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